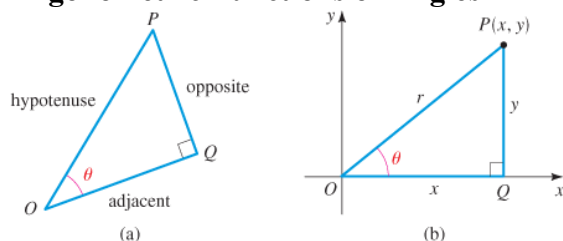


Chp 5.3: Trigonometric Functions of Angles**I. Trigonometric Functions of Angles****Definition of the Trigonometric Functions**

Let θ be an angle in standard position and let $P(x, y)$ be a point on the terminal side. If

$r = \sqrt{x^2 + y^2}$ is the distance from the origin to the point $P(x, y)$, then

$$\sin \theta = \frac{y}{r}$$

$$\cos \theta = \frac{x}{r}$$

$$\tan \theta = \frac{y}{x} \quad (x \neq 0)$$

$$\csc \theta = \frac{r}{y} \quad (y \neq 0)$$

$$\sec \theta = \frac{r}{x} \quad (x \neq 0)$$

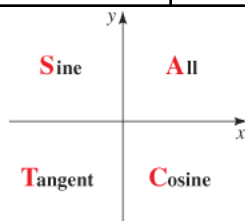
$$\cot \theta = \frac{x}{y} \quad (y \neq 0)$$

The angles for which the trigonometric functions may be undefined are the angles for which either the x - or y -coordinate of a point on the terminal side of the angle is 0. These angles are called **quadrantal angles**, that is, angles that are **coterminal** with the coordinate axes. It is a crucial fact that the values of the trigonometric functions **do not depend** on the choice of the point $P(x, y)$.

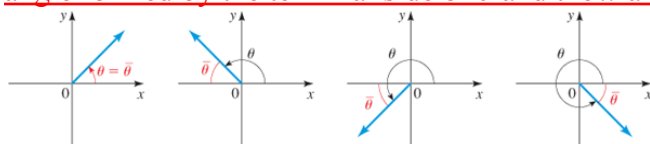
II. Evaluating Trigonometric Functions at Any Angle**Signs of the Trigonometric Functions**

Complete the following table to indicate which trigonometric functions are positive and which are negative in each quadrant.

Quadrant	Positive Functions	Negative Functions
I	<u>all</u>	<u>none</u>
II	<u>sin, csc</u>	<u>cos, sec, tan, cot</u>
III	<u>tan, cot</u>	<u>sin, csc, cos, sec</u>
IV	<u>cos, sec</u>	<u>sin, csc, tan, cot</u>

**Reference Angle**

Let θ be an angle in standard position. The reference angle $\bar{\theta}$ associated with θ is **the acute angle formed by the terminal side of θ and the x -axis**.



Evaluating Trigonometric Functions for Any Angle

To find the values of the trigonometric functions for any angle θ , we carry out the following steps

1. Find the reference angle $\bar{\theta}$ associated with the angle θ .
2. Determine the sign of the trigonometric function of θ by noting the quadrant in which θ lies.
3. The value of the trigonometric function of θ is the same, except possibly for sign, as the value of the trigonometric function of $\bar{\theta}$

III. Trigonometric Identities

The trigonometric functions of angles are related to each other through several important equations called trigonometric identities.

Fundamental Identities

Reciprocal Identities

$$\csc \theta = \frac{1}{\sin \theta}, \sec \theta = \frac{1}{\cos \theta}, \cot \theta = \frac{1}{\tan \theta}, \tan \theta = \frac{\sin \theta}{\cos \theta}, \cot \theta = \frac{\cos \theta}{\sin \theta}$$

Pythagorean Identities

$$\sin^2 \theta + \cos^2 \theta = 1 \qquad \tan^2 \theta + 1 = \sec^2 \theta \qquad 1 + \cot^2 \theta = \csc^2 \theta$$

IV. Areas of Triangles

The area A of a triangle with sides of lengths a and b and with included angle θ is

$$A = \frac{1}{2}ab \sin \theta$$

Example 1: Find the reference angle for the given angle.

- a. -199° b. $\frac{10\pi}{9}$ c. -1.4π d. 2.3 e. 485°

Example 2: Find the exact value of the trigonometric function.

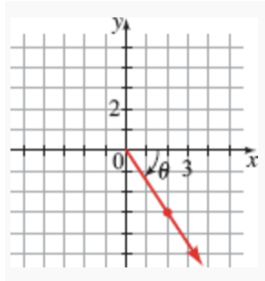
- a. $\csc 135^\circ$ b. $\sec -420^\circ$ c. $\tan -\frac{4\pi}{3}$ d. $\cos \frac{7\pi}{4}$ e. $\sin \frac{11\pi}{6}$

Example 3: From the information given, find the quadrant that θ lies in.

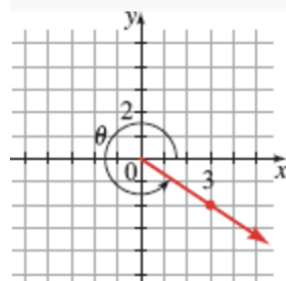
- a. $\tan \theta < 0$ and $\sin \theta < 0$ b. $\csc \theta > 0$ and $\cos \theta < 0$

Example 4: An angle θ is given in the coordinate plane. Find the exact value of the indicated trigonometric function of θ . (You may find it helpful to use the Pythagorean Theorem.)

a. $\cot \theta$



b. $\sec \theta$



Example 5: Find the values of the trigonometric functions of θ from the information given.

a. $\cos \theta = \frac{-3}{5}$, θ in Quadrant II

b. $\cot \theta = -2$, θ in Quadrant IV

c. $\csc \theta = -4$, $\tan \theta < 0$

d. $\sin \theta = \frac{3}{10}$, $\sec \theta > 0$

Example 6: Write the first trigonometric function in terms of the second for θ in the given quadrant.

a. $\sec \theta$, $\tan \theta$, Quadrant II

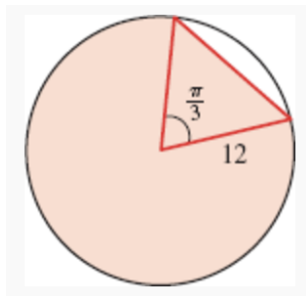
b. $\csc \theta$, $\cot \theta$, Quadrant III

Example 7: If $\theta = \frac{\pi}{3}$ find the value of each expression.

$\sin^2 \theta$, $\sin(\theta^2)$

Example 8:

- A triangle with sides of length 10 and 22 and included angle 10° . Find the area of the triangle.
- A triangle has area 16 in^2 . The angle between two sides of the triangle is 36° , where one of these two sides has length 5 in. Find the length of the other side.
- Find the area of the shaded region in the figure.



Example 9: A rocket fired straight upward is tracked by an observer on the ground, one mile away.

(a) Show that when the angle of elevation is θ , the height of the rocket (in feet) is $h = 5280 \tan \theta$.

(b) Complete the table to find the height of the rocket at the given angles of elevation.

θ	20°	60°	80°	85°
h				

